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Locating Places on the Earth – I

NCF features

In this lesson

- Art-integrated learning
- Map skills-based questions
- Scientific temper
- Observation task

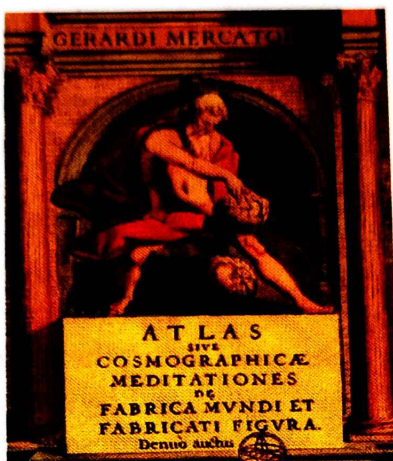
We know that it is impossible to see the entire Earth at the same time. This is due to the big size and spherical shape of the Earth. Even photographs taken by satellites show only half the Earth. Therefore, a model of the Earth can help us to view the entire Earth.

A small human-made model of the Earth is called a globe. A globe is a simple and accurate way of representing the Earth. The maps on the globe show the correct size, shape and location of the places in relation to each other. However, a globe has the following limitations:

- ◆ It is difficult to carry a globe.
- ◆ The space on the globe is limited and details cannot be shown on it.
- ◆ A globe cannot be made for a part of the Earth.
- ◆ Details of the geographic features cannot be shown properly.
- ◆ It is not only difficult to make a big size globe, but also difficult to handle it.

We need something simple and easy to learn about the Earth on which we live. Thus, we need maps drawn on a flat surface. However, it is not easy to draw maps on a flat surface.

Imagine cutting a globe



The picture of Atlas that appeared on the cover of Mercator's sixteenth century atlas.

and then flattening its pieces (See Fig. 1.1). It is not a continuous map and gives a distorted picture. It is not possible to make a correct map from a curved surface. Thus, we have developed some methods for drawing maps on flat surfaces. Try to cover a ball with a sheet of paper without making folds. You will not be able to do it. The same problem is faced when the spherical surface of the Earth is represented on a flat paper.

A map is the representation of the Earth as a whole or a part of it, drawn to a **scale** on a flat surface. Although less accurate than a globe, a map is more useful.

The word map has been derived from the Latin word 'mappo' meaning napkin. Mercator and Hondt were the first to publish a collection of maps in the form of a book. Its cover had the picture of the Greek mythological giant Atlas, holding the Earth. Since then a book of maps is called an **atlas**.

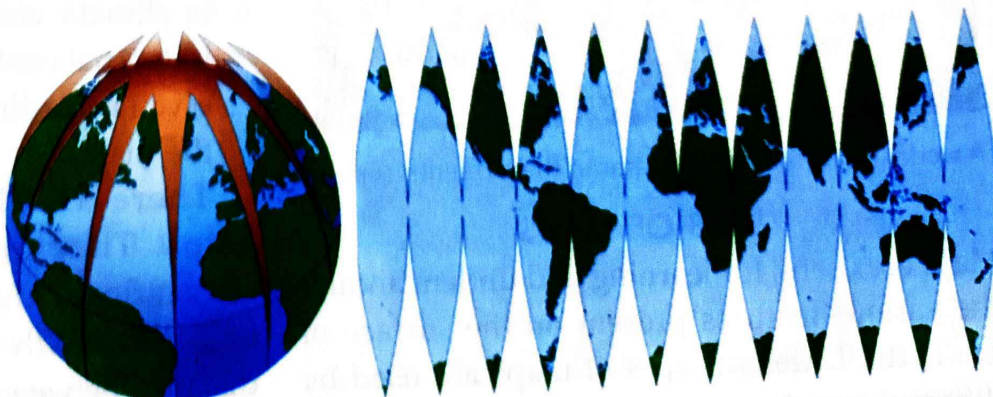
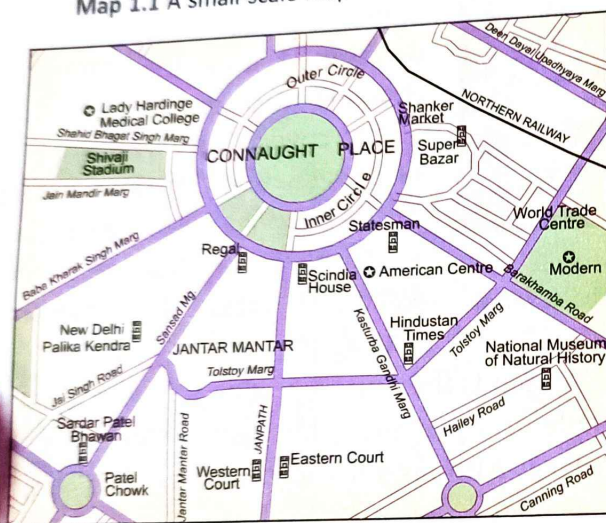


Fig. 1.1 A globe cut into parts and flattened into a map



Map 1.1 A small-scale map shows a large area.



Map 1.2 A large-scale map shows details of a small area.



A world map drawn by Mercator in the sixteenth century

TYPES OF MAPS

Maps are useful for learning and understanding the various features present on the surface of the Earth. Different types of maps are used by different people according to their needs.

Maps can be classified in many ways, but two of them are most useful. One classification is based on the scale of the map and the other is based on the function of the map.

According to scale, there are two types of maps – small-scale maps and large-scale maps.

- ◆ **Small-scale maps** show a very large area with less details. The maps in an atlas or the wall maps used in classrooms are small scale maps (See Map 1.1).
- ◆ **Large-scale maps** show a very small area in detail. The layout maps of villages and the guide maps of cities are large scale maps (See Map 1.2).

According to function, there are three main types of maps – physical maps, political maps and thematic maps.

- ◆ **Physical maps** show the relief features present on the Earth's surface like mountains, hills, plateaus, plains, peaks, valleys, rivers, oceans, etc. These maps help us to find suitable sites for human settlement, for building roads and railways, for the construction of dams, etc.
- ◆ **Political maps** show the boundaries between different countries and the boundaries of different political units within a country. These maps are used as base maps for showing various cultural, historical and economic activities.
- ◆ **Thematic maps** show specific type of information, such as roads, railways, air routes, distribution of population, conditions of climatic elements, location of mining and industrial centres, etc. The title of the map is given according to the information given on the map.

There are three main components of maps. They are distance, direction and **conventional signs** or symbols. The direction of a place with respect to the north and the distance between any two places are essential features of a map.

MEASURING DISTANCES

Distance means the space between any two places. Distances can be long or short. We normally say that the market is near our house and the school is far-away. Near means that we can reach a place in a short time, while far-away means that we need more time to reach that place. Thus, distance is related to the time taken to reach a destination. However, the time taken to cover a given distance varies not only from person to person but also from place to place. Thus, we need a system to measure distances accurately.

In India, we follow the metric system, in which long distances are measured in kilometres (km) and short distances in metres (m). The distance between any two points on a map is measured in centimetres (cm). We normally measure the straight-line distance.

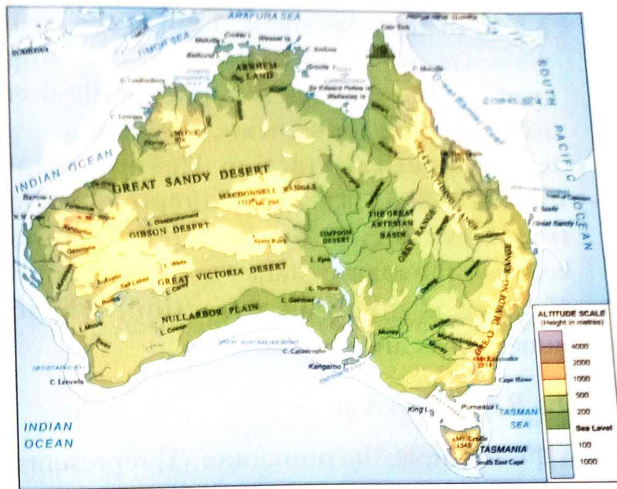
- ◆ We measure the distance between any two places on the map and relate it to the distance between the same two places on the ground.
- ◆ The distance between any two places on a map, measured along a straight line, is called the **map distance**.
- ◆ The distance between the same two points on the ground, measured along a straight line, is called the **ground distance**.
- ◆ The ratio between the map distance and the ground distance is called the **scale** of the map.

Suppose, the scale of a map is 1 cm : 1 km. This means that a distance of 1 cm between two points on the map corresponds to a distance of 1 km between the same two places on the ground.

How to show scale on a map?

The scale on a map can be shown in the following ways:

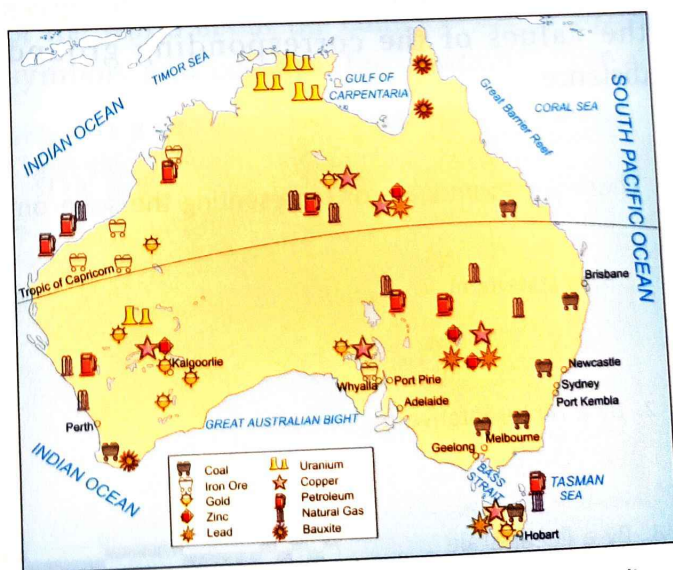
- ◆ **By a Statement** In the statement method, the scale is given in words, such as one centimetre is to two kilometres or 1 cm :



Map 1.3 Physical map of Australia



Map 1.4 Political map of Australia



Map 1.5 Thematic map showing minerals in Australia

2 km. In this statement, 1 cm is the map distance and 2 km is the corresponding ground distance. This is a quick method of using the scale for showing distances.

- ♦ **By a Numerical Fraction** The numerical fraction method is also called **representative fraction** or **R.F.** In this method, the ratio between the map distance and the ground distance is given as a fraction, whose numerator is always one.

The R.F. is given as $\frac{1}{1,00,000}$

In this example, the numerator (1) represents the map distance and the denominator (1,00,000) represents the ground distance.

$$\text{Thus, R.F.} = \frac{\text{Map distance}}{\text{Ground distance}}$$

No particular units of measurement are used with R.F. It has the universal application in map making.

- ♦ **By a Graphic or Linear Scale** In the graphic or linear scale method, the map distance is shown with the help of a straight line. The line is divided into parts called **primary divisions**. The first primary division is further subdivided into **secondary divisions**.

The linear scale helps to show the distance directly from the map. It also gives the values of the corresponding ground distance.

FINDING DIRECTIONS

The direction is shown on a map with the help of an arrow. Its tip is marked with the letter N. This means that the arrow is pointing towards the north. The arrow on a map is called the **North Line**.

The directions on the map or on the ground are always shown with respect to the north. If we know the **north**, then it is easy to find the other directions, namely **south**, **east** and **west**. These are the four cardinal directions on a map.

Once the four main directions are marked, we can easily find the intermediate directions. The direction midway between north and east is called the north-east (NE). Similarly, the other three intermediate directions are south-east (SE), north-west (NW) and south-west (SW) (See Fig. 1.2).

How to find the direction?

An easy and quick way to find the direction is by facing the rising Sun or with the help of the Pole Star.

- ♦ We know that the Sun rises in the east and sets in the west. If we stand facing the Sun in the morning, then we face the east. The west is towards our back. The left hand points towards the north and the right hand points towards the south (See Fig. 1.3).
- ♦ A magnetic compass is also used to find the north. The needle of the magnetic compass

There are three ways of representing the scale on a map.

1. By a statement

1 cm : 1 km

2. By a representative fraction (R.F.)

$\frac{1}{1,00,000}$

3. By a linear scale

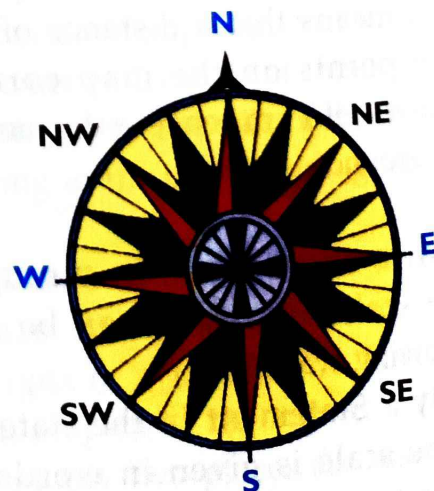
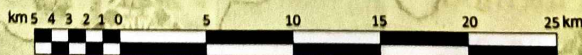


Fig. 1.2 Directions on a compass

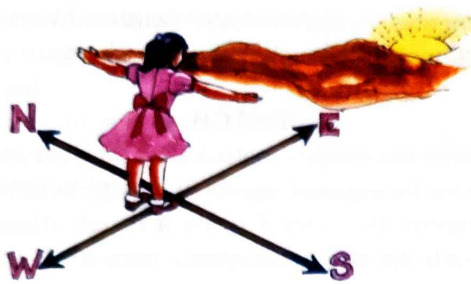


Fig. 1.3 Finding directions with the help of the rising Sun

always points towards the north-south direction. This compass is also called the **mariner's compass**, as it is used by mariners or seamen to find the direction while sailing.

- ♦ The Pole Star is located vertically above the North Pole. In the Northern Hemisphere, the constellation known as the Great Bear (called the Saptarishi in India) keeps revolving around the Pole Star. The two pointer stars of the Great Bear always point towards the Pole Star (See Fig. 1.4).

CONVENTIONAL SIGNS AND SYMBOLS

Maps are used for recording and showing a variety of information, such as political boundaries, relief features, climatic conditions, distribution of resources, population, means of transport, social and cultural features, cities and towns, etc.

There is not enough space on a map to show the actual size and shape of features such as

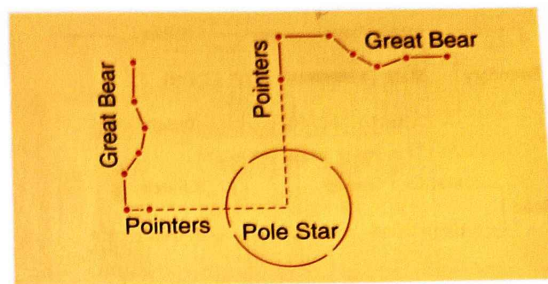


Fig. 1.4 Finding directions with the help of the Pole Star

If north is not shown on a map, the top of the map is taken as the north and the bottom as the south. When you stand facing the map, the right-hand side is the east and the left-hand side is the west.

mountains, peaks, rivers, lakes, bridges, dams, forts, forests, temples, churches, railway tracks and airports. However, a map without these features will be less informative. Thus, different colours, signs, symbols and letterings are used to present the information on the map. To understand the information given on a map, it is necessary to learn and follow the signs and symbols used in it. These signs and symbols keep the map clean and thus, makes it easy to read.

There is an agreement among many countries of the world about the common use of signs, symbols and colours. For example, blue is



A mariner's compass

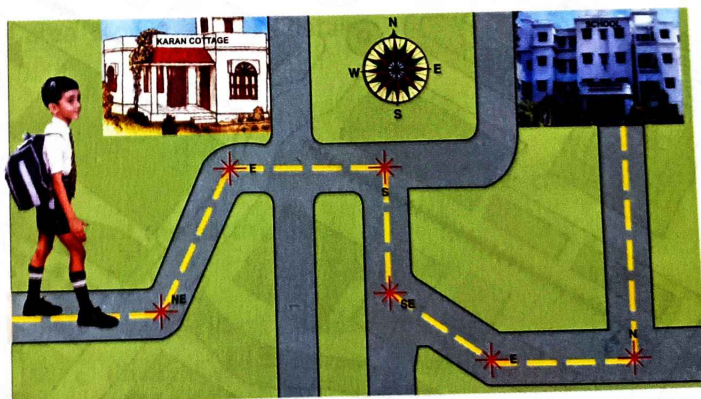


Fig. 1.5 The above diagram shows the directions that Karan should take while going to school. The compass points to the north.

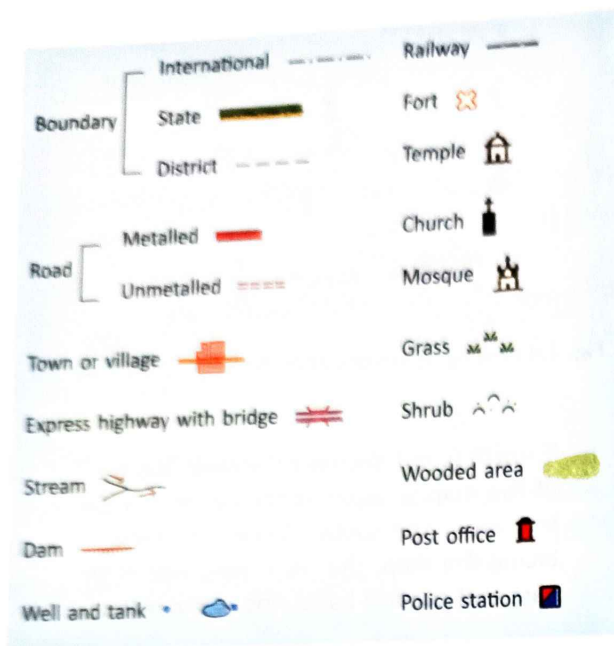


Fig. 1.6 Some conventional signs and symbols used on a map

always used to show water bodies and green is used for the plains. Thus, these signs and symbols are called conventional signs and symbols (See Fig. 1.6).

The information shown on a map can only be read and understood by those who know

its language, which includes features like scale, directions, colours, signs and symbols.

SKETCH

Suppose we want to help a new student in the class to locate and reach the neighbourhood bookshop. We would draw a rough diagram of roads to show where to turn and which landmark to look out for in order to reach the bookshop. With this rough diagram, the new student can reach the bookshop easily, even though the exact distance and directions are not mentioned. Such a diagram, which is drawn without measuring the distance between the various places, is called a **sketch**.

Thus, a sketch is a rough map with outlines, drawn without a scale and also without the proper use of signs and symbols.

PLAN

Sometimes, we want to draw an accurate map of a very small area like that of a house or a building. For example, if you and your family want to shift to a new house, it would be useful to have an accurate map of the house. This is

also called a **layout plan**.

This layout plan will show the length and the breadth of each room in the house, the height of windows and doors, etc. Such an accurate map of a very small area, which shows even the minor details, is called a **plan**. A plan is always drawn according to scale.

The scale of a plan is different from that of a map. Generally, a map represents a large area. So, it is drawn on a small scale. 1 cm on a map may represent 100 km on the ground. On the other hand, a plan represents a very small area, and is therefore

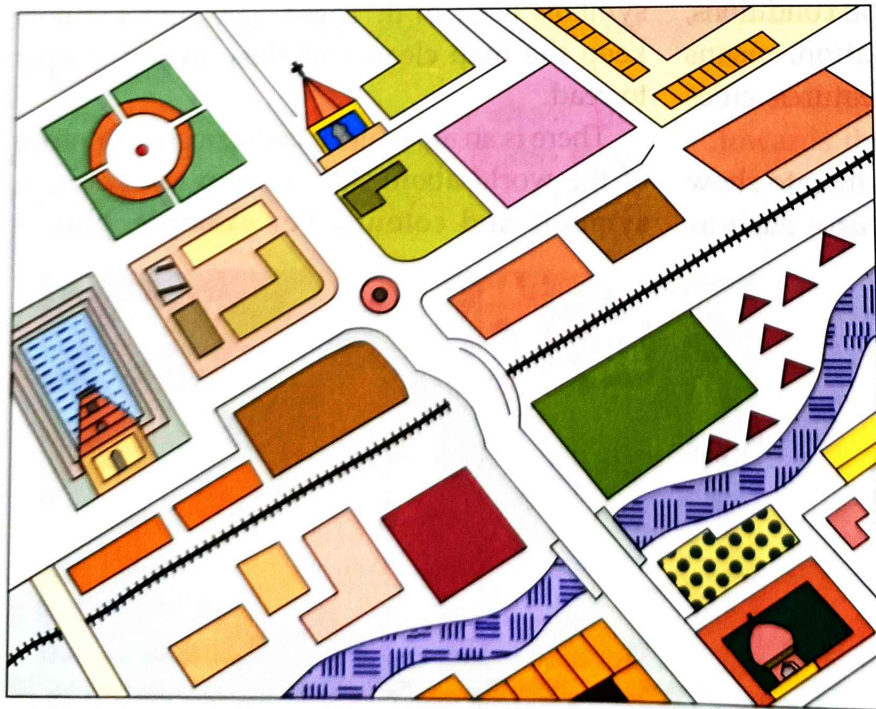


Fig. 1.7 A sketch of a neighbourhood

drawn on a large scale. 1 cm on a plan may represent just 1 m on the ground.

Thus, to sum up we can say:

- A map shows either a large area without details or a small area in detail.
- A plan shows a very small area in great detail.
- A sketch is a rough diagram drawn without a scale.

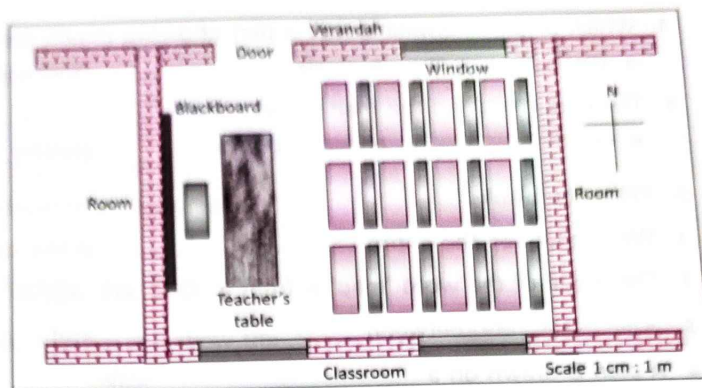


Fig. 1.8 A plan of a classroom

To Sum Up

- A globe is a simple and accurate way of representing the Earth.
- It is not possible to draw a correct map from a curved surface.
- Distance means the space between any two places.
- The distance between any two places on a map is called the map distance.
- The distance between any two places on the ground is called the ground distance.
- Scale is the ratio between the map distance and the ground distance.
- Scale is shown on the map in three ways—by a statement, by a representative fraction (R.F.) and by a linear scale.
- The directions on a map or on the ground is always shown with respect to the north.
- The four cardinal directions are north, south, east and west.
- Directions can be found with the help of the rising Sun, the Pole Star and the magnetic compass.
- Different colours, signs, symbols, letterings, etc., are used to show the information on a map.
- The standard symbols used universally are called conventional symbols.
- A map without a scale is called a sketch.
- A plan is a large-scale map of a very small area showing minute details.

Time to Learn

A. Tick (✓) the correct answers.

Multiple-choice questions

- Which of the following is the true representation of the Earth?
a. Map b. Globe c. Plan d. Sketch
- Which of the following maps shows a very large area with less details?
a. Large-scale b. Thematic c. Physical d. Small-scale
- The maps showing distribution of crops are _____ maps.
a. small-scale b. physical c. thematic d. political

4. Which of the following is **NOT** a part of a map drawn on a flat surface?
- Globe
 - Scale
 - Directions
 - Conventional symbols
5. The blue colour on the map shows
- clouds.
 - water bodies.
 - glaciers.
 - plains.

B. Fill in the blanks.

- We cannot see the entire _____ at the same time.
- The origin of the word 'map' is from a _____ word.
- Scale is the ratio between _____ and _____.
- A plan is drawn on a _____ scale.

C. Match the columns.

Column A

- Small-scale map
- Large-scale map
- Physical map
- Thematic map

Column B

- Guide maps
- Roads and railways
- Wall maps
- Mountains and rivers

D. Distinguish between the following.

- Globe and map
- Large-scale map and small-scale map
- Physical map and thematic map
- Map distance and ground distance

E. Explain the following terms.

- | | | | |
|-------------|----------|------------------------|-----------|
| 1. Map | 3. Scale | 5. Direction | 7. Sketch |
| 2. Distance | 4. R.F. | 6. Cardinal directions | 8. Plan |

F. Give very short answers for the following questions.

- What is a human-made model of the Earth called?
- What is a map showing the boundaries between different countries called?
- Name the system to measure distance followed in India.
- How is direction shown on a map?
- What is the Great Bear constellation called in India?

G. Give short answers for the following questions.

- Why do we need maps?
- How are maps more useful than a globe?
- What are the various components of a map?
- How can the Pole Star help in finding direction?

H. Give long answers for the following questions.

- What are the main limitations of a globe?
- Explain the purpose of different types of maps.

3. Describe the different methods of showing scale on a map.
4. How can we locate directions on a map?
5. What are the advantages of a mariner's compass?
6. Why is it not possible to draw an accurate map?
7. Why does R.F. have the universal application in map making?

HOTS

Time to Do

- I. Draw a sketch of your school building and complex and show the important places on it.

Art-integrated learning

- J. Study Map 1.5 in the chapter and answer the following questions.

Map skills-based questions

1. What does ★ show?
2. Which minerals are found in the central part of Australia?
3. Name the chief mineral found along the south-east coast of Australia.
4. Name the minerals found around the Gulf of Carpentaria.
5. Name the mineral found offshore along the west coast of Australia.

- K. Look at a globe and find out which longitudes are diametrically opposite to the given longitudes. Then, find the countries and oceans / seas which lie on the given longitudes and the countries and oceans / seas which lie on the diametrically opposite longitudes.

Scientific temper

1. 10° E
2. 90° W
3. 120° W
4. 45° E
5. 180° W
6. 65° E

Self-Assessment Checklist

	YES	NO	PARTLY
I found the longitudes opposite to the given longitudes.			
I identified the countries and oceans and seas on the given longitudes.			
I located the countries and ocean and seas which lie on the diametrically opposite longitudes to the given ones.			

- L. Get up early in the morning. Go to an open space near your house or your terrace during sunrise. Find the directions around your house with the help of the rising Sun.

Awareness of the immediate environment; Observation task

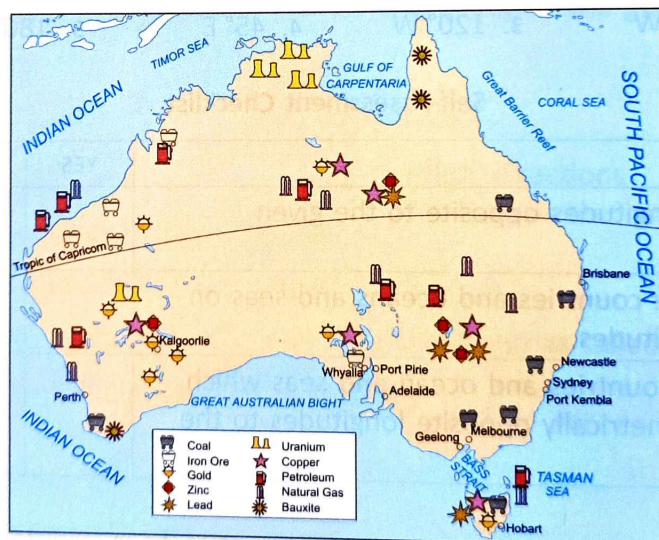
Assertion-Reason Questions

Two statements are given below. They are Assertion (A) and Reason (R).
Read both the statements and choose the correct option.

- Assertion (A): A model of the Earth can help us to view the entire Earth.
Reason (R): Due to the big size and spherical shape of the Earth, photographs taken by satellites show only half the Earth.
 - Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.
- Assertion (A): It is easy to find the direction by facing the rising Sun.
Reason (R): If we stand facing the Sun in the morning, then we face the west.
 - Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.

Map skills-based Questions

Look at the given map carefully. Tick the correct options.



- Which type of map is this?
 - Physical
 - Political
 - Thematic
 - Large-scale
- We get information about the distribution of _____ from the above map.
 - population
 - minerals
 - crops
 - forests
- The Pacific Ocean lies to the east of Australia, whereas the Indian Ocean lies to the _____ of the continent.
 - north
 - central
 - south
 - west
- Which of these is **NOT** shown in the map?
 - Manganese
 - Iron ore
 - Petroleum
 - Gold